

Computational Thinking

Discrete Mathematics

Number Theory

Topic 02 : Logic

Logic

Lecture 01 : Introduction to Propositional Logic

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Graphs and
Networks

Autumn Semester, 2026/27

Collections

Outline

- Propositions and fundamental logical operators (AND, OR and NOT).
- Evaluating logical expression using truth tables.
- Satisfiability, Tautologies and Contradictions.

Enumeration

Relations & Functions

Thought for the day ...

While walking through a fictional forest, you encounter three identical trolls guarding a bridge. Each troll is either a knight, who always tells the truth, or a knave, who always lies. The trolls will not let you pass until you correctly identify each as either a knight or a knave. Each troll makes a single statement:

If I am a knave, then there are exactly two knights here.



Troll 1 is lying.



Either we are all knaves or at least one of us is a knight..



Which troll are knights? and which are knaves?

1. Introduction

3

- Propositional logic is concerned with analysing propositions (true or false statements).
- A proposition may be atomic or compound (build up using logical connectives).
- Constructing compound propositions using *And*, *Or* and *Not*.

2. Truth tables

12

- Evaluating an expression for all possible input combinations.

3. Tautologies and Contradictions

20

- Statements that are always true or always false.

Logic

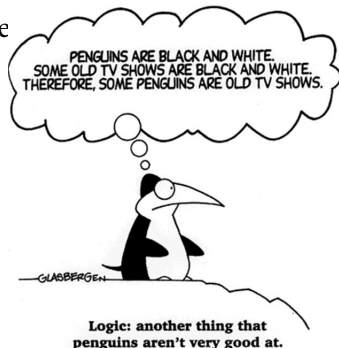
Logic is “science of reasoning”

- Allows us to represent knowledge in precise, unambiguous way.
- Allows us to make valid inferences using a set of consistent rule
- Roots of logic date back to the ancient Greeks, e.g., Aristotle.
- Greeks were interested in valid logical inference rules, such as syllogisms:

“All men are mortal.

Socrates is a man.

Therefore, Socrates is mortal.”



The Partially Examined Life podcast: www.partiallyexaminedlife.com

The Fallacy-a-Day Podcast: <http://fallacyaday.com>

Propositional Logic

- The building blocks of propositional logic are propositions

Definition 1 (Proposition)

A **proposition** (**statement**) is a sentence that is either **True** or **False**.

- Examples:

“Java is a programming language.”

True

“Cork is the capital of Ireland.”

False

“ $1 + 2 = 3$ ”

True

“Today is Tuesday.”

depends

“The universe is fine-tuned.”

unknown (at present)

- Examples of sentences that are not propositions/statements:

- “How are you?”*

— A question cannot be assign a **True/False** value.

- “Stop sleeping in class!”*

— An order cannot be assign a **True/False** value.

- “Correct horse battery staple.”*

— Not a sentence.

- “This sentence is false.”*

— Pathological example.

Propositional Variables, Truth Value

Given a proposition we are interested in knowing its **truth value**.

Definition 2 (Truth Value)

The **truth value** of a proposition identifies whether a proposition is false (written **False** or **F** or 0) or true (written **True** or **T** or 1).

Question

What is truth value of “*Tuesday is the day after Sunday*” ?

F

Notation

- Variables that represent propositions are called propositional variables.
- Denote propositional variables using lower-case letters, such as $p, p_1, p_2, q, r, s, \dots$
- Truth value of a propositional variable is either **T** or **F**.

Compound vs Atomic Propositions

- Propositional logic allows constructing more complex propositions from atomic ones.
- More complex propositions formed using **logical connectives** (also called **boolean connectives** or **logical operators**).
- The three basic logical connectives:

Connective	Symbol	Python
conjunction (AND)	$\wedge$	<code>and</code>
disjunction (OR)	$\vee$	<code>or</code>
negation (NOT)	$\neg$	<code>not</code>

- Propositions formed using these logical connectives called **compound propositions**; otherwise called **atomic propositions**.

Today is wet and I am hungry

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- Propositions formed using these logical connectives called **compound propositions**; otherwise called **atomic propositions**.

Today is wet and I am hungry
 { atomic } { atomic }
 { compound }

Exercise

Classify each of the sentences below as an atomic statement, a compound statement, or not a statement at all.

- ① The sum of the first 100 odd positive integers.
- ② Everybody needs somebody sometime.
- ③ Waterford will win the All-Ireland or I'll eat my hat.
- ④ Go to your room!
- ⑤ Every natural number greater than 1 is either prime or composite.
- ⑥ This sentence is false.

Exercise

Classify each of the sentences below as an atomic statement, a compound statement, or not a statement at all.

① The sum of the first 100 odd positive integers.

—*This is not even a sentence (no verb).*

② Everybody needs somebody sometime.

—*This is an atomic statement.*

③ Waterford will win the All-Ireland or I'll eat my hat.

—*This is a compound statement.*

④ Go to your room!

—*This is an order, not a statement*

⑤ Every natural number greater than 1 is either prime or composite.

—*This is a compound statement.*

⑥ This sentence is false.

—*This is sentence but is not a statement.*

Conjunction (AND)

- **Conjunction** of two propositions, p and q , written as $p \wedge q$, is the proposition:
“ p and q ”
- What is the relationship between the truth value of p and of q and the truth value of $p \wedge q$?

$$p \wedge q = \begin{cases} \mathbf{T} & \text{if both } p \text{ is } \mathbf{T} \text{ and } q \text{ is } \mathbf{T} \\ \mathbf{F} & \text{otherwise} \end{cases}$$

Example

What is the conjunction and the truth value of $p \wedge q$ for ...

- $p = \text{“It is a autumn semester”}$, $q = \text{“Today is Thursday”}$
- $p = \text{“It is Tuesday”}$, $q = \text{“It is morning”}$

Disjunction (OR)

- **Disjunction** of two propositions, p and q , written as $p \vee q$, is the proposition
“ p or q ”
- What is the relationship between the truth value of p and of q and the truth value of $p \vee q$?

$$p \vee q = \begin{cases} \mathbf{T} & \text{if either } p \text{ is } \mathbf{T} \text{ or } q \text{ is } \mathbf{T}, \text{ or both are } \mathbf{T} \\ \mathbf{F} & \text{otherwise} \end{cases}$$

Example

What is the disjunction and the truth value of $p \vee q$ for ...

- $p = \text{“It is a autumn semester”}$, $q = \text{“Today is Thursday”}$
- $p = \text{“It is Friday”}$, $q = \text{“It is morning”}$

Negation (NOT)

- **Negation** of a proposition, p , written, $\neg p$, represents the proposition:
“It is not the case that p .”
- What is the relationship between the truth value of p and $\neg p$?

If p is **T**, then $\neg p$ is **F** and vice versa.

- In simple English, what is $\neg p$ if p stands for ...

p	$\neg p$
“Today is Tuesday.”	“Today is not Tuesday.”
“ $1 + 1 = 2$ ”	“ $1 + 1 \neq 2$ ”

- Properties of NOT
 - $\neg \neg p = p$

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2. Truth tables

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- Evaluating an expression for all possible input combinations.

3. Tautologies and Contradictions

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- Statements that are always true or always false.

Propositional Formulas and Truth Tables

- A **propositional formula** is logical expression constructed from atomic and compound propositions and logical connectives.
- A **truth table** for a propositional formula, A , shows the truth value of A for every possible value of its constituent atomic propositions.

Not / Negation

p	$\neg p$
F	T
T	F



NOT

And / Conjunction

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T



AND

Or / Disjunction

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T



OR

Truth tables and Logic Gates

AND

NOT

OR

p	$\neg p$

p	q	$p \wedge q$

p	q	$p \vee q$

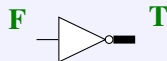
Truth tables and Logic Gates

AND

NOT

OR

p	$\neg p$
F	T



p	q	$p \wedge q$

p	q	$p \vee q$

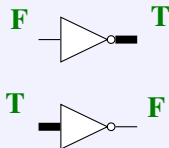
Truth tables and Logic Gates

AND

NOT

OR

p	$\neg p$
F	T
T	F

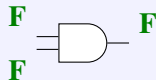


p	q	$p \wedge q$

p	q	$p \vee q$

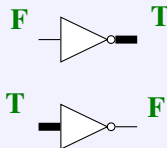
Truth tables and Logic Gates

AND

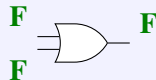


NOT

p	$\neg p$
F	T
T	F



OR

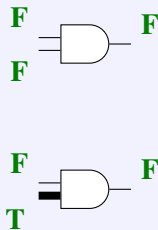


p	q	$p \wedge q$
F	F	F

p	q	$p \vee q$
F	F	F

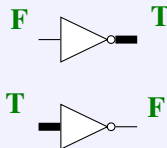
Truth tables and Logic Gates

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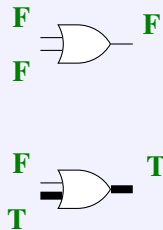


NOT

p	$\neg p$
F	T
T	F



OR

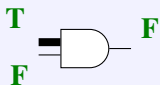
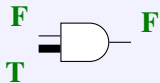
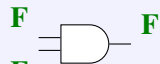


p	q	$p \wedge q$
F	F	F
F	T	F

p	q	$p \vee q$
F	F	F
F	T	T

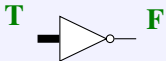
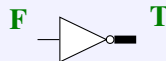
Truth tables and Logic Gates

AND



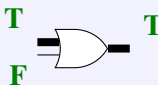
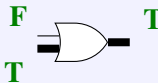
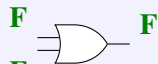
p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F

NOT



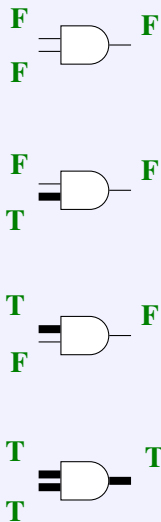
p	q	$p \vee q$
F	F	F
F	T	T
T	F	T

OR



Truth tables and Logic Gates

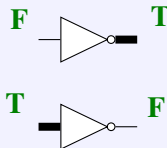
AND



p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

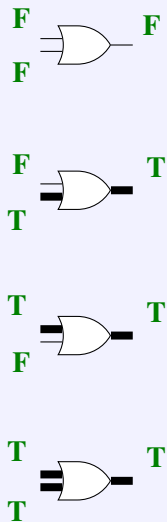
NOT

p	$\neg p$
F	T
T	F



p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

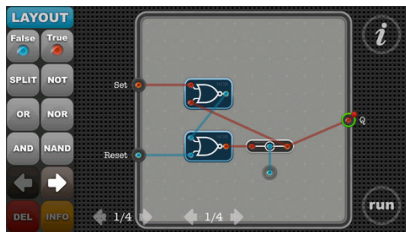
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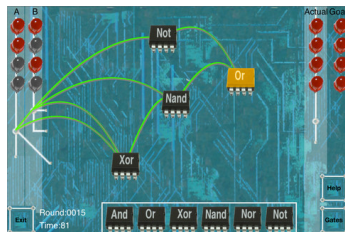
Other Resources

iPad/iPhone Apps (assume similar on Android)

Circuit Coder



Boolean Master



Videos

- <https://class.coursera.org/cs101/lecture/17>
Part of the Computer Science 101 by Nick Parlante on coursera.

Constructing Truth Tables

Useful strategy for constructing truth tables for a formula:

- STEP 1 Identify the constituent atomic propositions of the formula.
- STEP 2 Identify compound propositions in within the formula in increasing order of complexity, including the formula itself.
- STEP 3 Construct a table enumerating all combinations of truth values for atomic propositions.
- STEP 4 Fill in values of compound propositions for each row.

Examples

Construct truth tables for the following formulas:

- 1 $(p \vee q) \wedge \neg p$
- 2 $(p \wedge q) \vee (\neg p \wedge \neg q)$
- 3 $(p \vee q \vee \neg r) \wedge r$

Example 1: $(p \vee q) \wedge \neg p$

- STEP 1 Identify the constituent atomic propositions ... p and q
- STEP 2 Identify compound propositions ...
- STEP 3 Enumerate all combinations of truth values for atomic propositions ...
- STEP 4 Fill in values of compound propositions for each row ...

Example 1: $(p \vee q) \wedge \neg p$

STEP 1 Identify the constituent atomic propositions ... p and q

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STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q

Example 1: $(p \vee q) \wedge \neg p$

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STEP 4 Fill in values of compound propositions for each row ...

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$

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- STEP 2** Identify compound propositions ...
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- STEP 4** Fill in values of compound propositions for each row ...

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F			
F	T			
T	F			
T	T			

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p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F	F	T	F
F	T			
T	F			
T	T			

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- STEP 3** Enumerate all combinations of truth values for atomic propositions ...
- STEP 4** Fill in values of compound propositions for each row ...

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F	F	T	F
F	T	T	T	T
T	F			
T	T			

Example 1: $(p \vee q) \wedge \neg p$

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p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F	F	T	F
F	T	T	T	T
T	F	T	F	F
T	T			

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p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F	F	T	F
F	T	T	T	T
T	F	T	F	F
T	T	T	F	F

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- STEP 3** Enumerate all combinations of truth values for atomic propositions ...
- STEP 4** Fill in values of compound propositions for each row ...

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
F	F	F	T	F
F	T	T	T	T
T	F	T	F	F
T	T	T	F	F

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

- STEP 1 Identify the constituent atomic propositions ... p and q
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Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

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p q

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p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$

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p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F					
F	T					
T	F					
T	T					

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

STEP 1 Identify the constituent atomic propositions ... p and q

STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F	F	T	T	T	T
F	T					
T	F					
T	T					

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

STEP 1 Identify the constituent atomic propositions ... p and q

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STEP 4 Fill in values of compound propositions for each row ...

p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F	F	T	T	T	T
F	T	F	T	F	F	F
T	F					
T	T					

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

STEP 1 Identify the constituent atomic propositions ... p and q

STEP 2 Identify compound propositions ...

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STEP 4 Fill in values of compound propositions for each row ...

p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F	F	T	T	T	T
F	T	F	T	F	F	F
T	F	F	F	T	F	F
T	T					

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

- STEP 1** Identify the constituent atomic propositions ... p and q
- STEP 2** Identify compound propositions ...
- STEP 3** Enumerate all combinations of truth values for atomic propositions ...
- STEP 4** Fill in values of compound propositions for each row ...

p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F	F	T	T	T	T
F	T	F	T	F	F	F
T	F	F	F	T	F	F
T	T	T	F	F	F	T

Example 2: $(p \wedge q) \vee (\neg p \wedge \neg q)$

- STEP 1** Identify the constituent atomic propositions ... p and q
- STEP 2** Identify compound propositions ...
- STEP 3** Enumerate all combinations of truth values for atomic propositions ...
- STEP 4** Fill in values of compound propositions for each row ...

p	q	$(p \wedge q)$	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
F	F	F	T	T	T	T
F	T	F	T	F	F	F
T	F	F	F	T	F	F
T	T	T	F	F	F	T

Example 3: $(p \vee q \vee \neg r) \wedge r$

- STEP 1 Identify the constituent atomic propositions ... p , q , and r
- STEP 2 Identify compound propositions ...
- STEP 3 Enumerate all combinations of truth values for atomic propositions ...
- STEP 4 Fill in values of compound propositions for each row ...

Example 3: $(p \vee q \vee \neg r) \wedge r$

STEP 1 Identify the constituent atomic propositions ... p , q , and r

STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q	r

Example 3: $(p \vee q \vee \neg r) \wedge r$

STEP 1 Identify the constituent atomic propositions ... p , q , and r

STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q	r	$\neg r$	$(p \vee q \vee \neg r)$	$(p \vee q \vee \neg r) \wedge r$

Example 3: $(p \vee q \vee \neg r) \wedge r$

STEP 1 Identify the constituent atomic propositions ... p , q , and r

STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q	r	$\neg r$	$(p \vee q \vee \neg r)$	$(p \vee q \vee \neg r) \wedge r$
F	F	F			
F	F	T			
F	T	F			
F	T	T			
T	F	F			
T	F	T			
T	T	F			
T	T	T			

Example 3: $(p \vee q \vee \neg r) \wedge r$

STEP 1 Identify the constituent atomic propositions ... p , q , and r

STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

p	q	r	$\neg r$	$(p \vee q \vee \neg r)$	$(p \vee q \vee \neg r) \wedge r$
F	F	F	T	T	F
F	F	T			
F	T	F			
F	T	T			
T	F	F			
T	F	T			
T	T	F			
T	T	T			

Example 3: $(p \vee q \vee \neg r) \wedge r$

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STEP 2 Identify compound propositions ...

STEP 3 Enumerate all combinations of truth values for atomic propositions ...

STEP 4 Fill in values of compound propositions for each row ...

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F	F	F	T	T	F
F	F	T	F	F	F
F	T	F			
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F	T	T	F	T	T
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F	F	T	F	F	F
F	T	F	T	T	F
F	T	T	F	T	T
T	F	F	T	T	F
T	F	T	F	T	T
T	T	F	T	T	F
T	T	T	F	T	T

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- Propositional logic is concerned with analysing propositions (true or false statements).
- A proposition may be atomic or compound (build up using logical connectives).
- Constructing compound propositions using *And*, *Or* and *Not*.

2. Truth tables

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- Evaluating an expression for all possible input combinations.

3. Tautologies and Contradictions

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- Statements that are always true or always false.

Introduction to Propositional Logic — Summary

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Satisfiable, Tautologies and Contradictions

Satisfiable

A proposition is **satisfiable** if it is **True** for at least one set of inputs (case).

Tautology

A **tautology** is an expression involving logical variables that is **True** in all cases.

- Examples

- $p \vee \neg p$
- $(p \wedge q) \vee (p \wedge \neg q) \vee \neg p$

“Tomorrow, I will be dead or I will be alive”

Contradiction

A **contradiction** is an expression involving logical variables that is **False** in all cases.

- Examples

- $p \wedge \neg p$

“On Friday, I will win the lottery and not win the lottery.”